PROJECT:

1X45MW CAPTIVE POWER PLANT,(PHASE-I) AT JAMSHEDPUR, JHARKHAND.

DOCUMENT TITLE:

TECHNICAL SPECIFICATION

FOR DC SYSTEM

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TABLE OF CONTENTS

1. INTRODUCTION 4

2. SCOPE OF SUPPLY 4

3. STANDARD 4

4. DESIGN BASIS 5

- **4.1.** INPUT PARAMETERS 5
- **4.2.** DESIGN BASIS & SIZING CRITERIA 5
- **4.3.** LAYOUT 5

5. DUAL FCBC 6

- **5.1.** SYSTEM REQUIREMENTS 6
- **5.2.** OPERATION PHILOSOPHY 6
- **5.3.** GENERAL REQUIREMENTS 6
- **5.4.** ACCESSORIES 8
- **5.5.** METERING AND INDICATION CIRCUIT 9
- **5.6.** ANNUNCIATION 10
- **5.7.** GROUNDING 11
- **5.8.** PAINTING 11
- **5.9.** TEST REQUIREMENTS 11

6. BATTERY 12

- **6.1.** GENERAL 12
- **6.2.** BATTERY ISOLATION BOX 13
- **6.3.** ACCESSORIES 13
- **6.4.** TESTS 13

7. DC DISTRIBUTION BOARD 14

- **7.1.** CONSTRUCTIONS 14
- **7.2.** INCOMER & BUS COUPLER MODULE 15
- **7.3.** BUS BARS 16
- **7.4.** OUTGOING FEEDER MODULE 17
- **7.5.** GROUNDING 17
- **7.6.** PAINTING 18

8. SPARES 18

9. DRAWINGS, DATA AND MANUALS 18

10. ANNEXURE 19

1. INTRODUCTION

This specification covers design, engineering, manufacture, shop testing, inspection, packing, transportation, delivery to site, unloading & storage at site, erection, testing and commissioning of One (1) sets of 220V D.C System with all accessories and suitable for site condition.

It is not the intent to specify completely all details of design and construction of the equipment. However, the equipment shall conform in all respects, to high standards of design, engineering and workmanship and be capable of performing in continuous commercial operations up to the Bidder's guarantee in a manner acceptable to the Owner/Consultant who will interpret the meaning of drawings and specifications and shall have power to reject any work or materials which in his judgment are not in full accordance therewith.

2. SCOPE OF SUPPLY

The scope of supply shall include but not limited to the following:

For Phase-1:-

- One (1) Set of 220V DC system shall consist of 2 x 100% dual float cum boost chargers associated with 2 X 100% rated batteries for STG Loads, 33kV Switchgear Panel and LT PCC panel loads of Power Plant and the same shall be designed for 10 hours backup.
- One (1) Set of DC Distribution Board.
- All Relevant Drawings, Data and Instruction Manuals.

Common:-

- One Set of Special Tools and Tackle.
- Mandatory Spares.

3. STANDARD

The design, manufacture and testing of equipment shall be carried out as per the latest Indian/International Standards, Indian Electricity Rules, Relevant Code of Practices and requirements of Chief Electrical Inspector of state electricity board.

Equipment/components conforming to the latest editions of Standards of the International Standard Institutions, Institution of Electrical & Electronic Engineers, USA (IEEE), National Elect. Manufactures Association, USA (NEMA) or International Electro-Technical Commission (IEC) will also be acceptable provided these are either equivalent or more stringent.

4. DESIGN BASIS

4.1. Input Parameters

Following auxiliary supplies with +/- ten (10) percent voltage variation, +/- five (5) percent frequency variation and ten (10) percent combined voltage and frequency variation are available:

Three (3) Phase, 415V, Four (4) Wire, 50 Hertz, effectively earthed AC supply.

Single phase 230V, 50Hz AC supply for cooler control, panel/cabinet space heating and illumination circuits.

In case offered equipment needs any other supply, it shall be arranged by the Contractor by providing suitable transformer, converter and/or inverter.

4.2. Design Basis & Sizing Criteria

The 220V battery and battery charger will constitute the D.C. Power supply. The batteries and chargers will be installed indoor in a clean but hot, humid and tropical atmosphere. The battery and charger combinations shall be such as to ensure continuity of D.C. supply at load terminals without even momentary interruption. For continuous operation at specified ratings, temperature rise of the various components of battery and charger shall be limited to the permissible values as stipulated in relevant standards. Further the voltage at load terminals shall never exceed the limits of +10% and -15% of the nominal system voltage. The batteries shall be so sized as to meet continuous load for 10 hrs and intermittent / momentary loads for 1 minute. While estimating battery capacities, a design margin of 10% shall be kept. For the purpose of estimating battery capacities, the maximum and minimum temperatures shall be considered as 27 °C and -2 °C respectively. All momentary loads shall be treated as one minute loads. The procedure for estimating battery capacities shall be as per guidelines stipulated in IEEE-485.

The battery will be sized such that the voltage at any time during the duty cycle shall not be less than 1.85 volt per cell.

The float-cum-boost charger shall be sized to restore the fully discharged battery to full charge condition in ten (10) hours with 20% margin over maximum charging rate or to operate as a float charger with duty requirement to cater the total continuous load and the trickle charging current of the battery plus a 20% margin. The batteries & chargers shall be so designed that the maximum fault level on DCDB is limited to 15kA.

4.3. Layout

The battery and battery charger shall be located inside the STG building. Sufficient clear space shall be provided for attending individual cells.

5. DUAL FCBC

5.1. System Requirements

Two sets of Dual Float cum Boost chargers (A & B) shall be considered to cater each power plant DC System.

Each Dual FCBC shall operate in parallel feeding the DC load and float charging the battery bank under normal operation. Each Dual FCBC shall have one number of Float cum Boost charging section with redundancy (i.e.) there will be two independent Float cum Boost charge unit, so that in case of outage of one unit, the other will take over bumplessly.

5.2. Operation Philosophy

The Dual FCBC's shall accomplish the following operating conditions without any disturbances in the DC System.

During Normal Operating Condition:-

- Dual FCBC-A supplies the DCDB Bus-A loads and float charges the Battery Bank-1.
- Dual FCBC-B supplies the DCDB Bus-B loads and float charges the Battery Bank-2.
- DCDB Bus Coupler breaker shall be kept OPEN.

During Dual FCBC Main Charger (in A or B) Failure Condition:-

Whenever main charger of one of the dual chargers (A or B) fail, the standby charger of the respective dual charger (A or B) will come into operation and supplies the load (DCDB Bus-A or Bus-B) and charges the battery without interruption to the load.

During Dual FCBC Both Main & Standby Charger Failure (A or B) Condition:-

Whenever main and standby chargers of one of the dual FCBC Chargers (A or B) fail, the other dual FCBC charger will come into operation. The DCDB incomer breaker of the failed dual FCBC charger shall be kept OPEN and bus coupler shall be CLOSED and vice versa.

During Incoming AC Power Supply Failure Condition:-

Whenever the power supply to both dual FCBC fails, the battery bank-1 supplies the DCDB Bus-A Load and battery bank-2 supplies the DCDB Bus-B Load.

5.3. General Requirements

The float-cum-boost charger will be normally operated as a float charger; will be normally ON, supplying the D.C. load and at the same time trickle charging the battery. The characteristics shall be such that if load is high and exceeds the charger capacity, the excess load will be supplied by the battery & in boost charging mode to boost charge the battery up to 2.75 Volts per cell.

The float cum boost charger shall be suitable for float charging as well as boost charging the battery. Each battery charger shall be capable of float charging the battery, while supplying

the power plant normal DC load. The design shall be such that, in case the load exceeds the charger capacity, the excess load current shall be supplied by the battery.

The other float cum boost charger will be normally in float mode and will cut in to the circuit automatically to take over the functions of other charger in case of its failure.

On failure of station A.C. supply, both float cum boost chargers will go out of service and battery will take over to supply station emergency loads automatically. In the event of float charger failure, when battery is boost charged by boost charger, continuity in DC supply to the load (DCDB) shall be maintained from the charger through dropper diodes to the load.

The process of change over from float to boost charging and from boost to float charging shall be selectable by an auto / manual selector switch. In auto mode, the changeover shall be initiated through a current sensor.

In manual mode, the changeover shall be performed manually using selector switches / push buttons. Manual switch shall have priority over automatic switching.

Suitable Potentiometers for Voltage and Current adjustment shall be provided under float mode & Boost mode.

When boost charging mode is selected, the battery shall be charged by constant current mode or by constant voltage mode as per battery manufacturer's recommendation. Selection of constant current mode or constant voltage mode shall be fully automatic.

A digital timer shall be provided for initiating the changeover to float mode by default after a preset time period.

In the event of failure of the charger feeding the load, when battery is being boost charged by the other charger, DC supply to the loads shall be maintained from the charger through dropper diode (if required) or through a contactor. Interlock shall be provided in each charger to disconnect the charger from both the DC load and the charger operating under float mode.

The battery charger shall be of solid state using silicon rectifiers with full wave, fully controlled bridge configuration and complete with automatic voltage regulator, current limiting circuitry, surge suppression network, smoothing filter circuits and soft-start feature. The battery charger shall have an isolator/MCCB at the incomer, to facilitate panel maintenance. The battery charger shall be suitable for installation and satisfactory operation in a pressurized or non-pressurized room with restricted natural air ventilation.

The battery charger shall be suitable for input power supply of 415 V ± 10%, 3 phase, 50 Hz ±5% incoming supply. The voltage regulation shall be within ± 1% for a (0-100)% load variation and +/-10% combined input AC voltage and frequency variation on the AC side. The battery charger shall be provided with 2 x 100% capacity cooling fans.

Each charger shall be sized for the most stringent of the following duty conditions, whichever is higher:

- The charger shall be sized considering off line boost charging current in Amps of 0.25 Ah (C10) of VRLA battery.
- With on line float charging of battery, the charger current in Amps shall be 1.20 x Average DC load + Float charging current. The loading conditions shall be as elaborated in battery specifications.
- Initial charging of uncharged battery assembly to fully charged condition

The charger shall be self-protecting against all AC and DC transients and steady state abnormal currents and voltages.

The charger shall be suitable for initial charging of uncharged battery assembly.

The charger shall be provided with blocking diodes to prevent reverse current flow into charger.

The charger shall be provided with dropper diode with bypass switch arrangement to prevent the DC output over voltage during boost charging.

Charger will have a separate over voltage relay with a variable setting from 100% to 120% in steps of 5% of Vn.

Filter circuits consisting of smoothing choke and condenser to limit the ripple content at the output to the maximum of 1/2 - 1% of the DC voltage.

All electronic power devices like Thyristor, transistors, diodes etc. shall be rated under operating conditions for at least 150% of the maximum current carried by the device.

All electrical components such as transformers, reactors, contactors, switches, bus bars etc. shall be rated for at least 125% of the maximum required rating. No electronic device shall experience a PIV greater than 50% of its rated value. The output voltage shall be restored within the steady state limits within 250 ms.

The maximum noise level from the charges measured at 1 meter distance in any position, at any load between (0-100)% shall not exceed 75 dB (A).

The DC Distribution Board in the battery chargers shall be provided as per the system requirement.

5.4. Accessories

The battery charger shall be equipped with, but not limited to the following;

MCCB on AC & DC side

Solid State Rectifier Circuits

Dry type, double wound with copper conductor rectifier transformers with class F insulation & temperature rise shall be limited to Class B.

Fully rated blocking diode with redundancy for reliability, so that failure of one diode will not affect the system.

Protection against reverse battery connection.

A high impedance earth fault relay shall be provided for the protection of the battery.

Silicon blocking diodes shall be provided in the charger output circuit to prevent back-feed from battery into the charger and filters.

Air break type incoming AC contactor with thermal overload relays.

Heavy duty, load-break type incoming isolating switches operated by a handle with padlocking facility in ON and OFF position.

HRC and fast acting semi conducting fuses with indication for float cum boost charger.

LED type indicating lamps.

Manual / Auto control for selection of float and boost charger.

Voltage setters for setting the output of float/ boost charger independently.

Current limit setters.

Alarm acknowledges, reset and test button.

Solid state surge protectors on both AC & DC side.

Lamps to indicate float/boost mode of operation.

5.5. Metering and Indication Circuit

Charger Panel metering and indication circuit shall include and not limited to the following;

Meters:

Charger-A1 & B1

- AC Input Voltmeter with Selector Switch
- AC Input Ammeter with Selector Switch
- DC Charger Output Voltmeter
- DC Charger Output Ammeter
- DC Voltmeter with Selector Switch (Battery Voltage)
- DC Ammeter (Battery)
- DC Earth Leakage Ammeter (mA).
- DC Load Voltmeter
- DC Load Ammeter

Charger-A2 & B2

- AC Input Voltmeter with Selector Switch
- AC Input Ammeter with Selector Switch
- DC Charger Output Voltmeter
- DC Charger Output Ammeter

Lamps:

- AC Power Supply (1 lamp for each phase)
- Mains ON
- Float Charger ON
- Boost Charger ON
- Charger Fault

In addition to above, provision shall be given for remote monitoring at DCS through suitable transducers of galvanically isolated with 4-20mA dual outputs for the following:

- Battery Voltage Low, High and Normal
- DC Voltage of Float & Boost Charger
- DC Current of Float & Boost Charger

5.6. Annunciation

The following annunciations shall be provided and also potential free contacts for the following shall be made available for hooking up from annunciation panel to DCS control panel:

- SCR Fuse Failure
- Charger Overload
- Charger Fail/Trip
- Battery Fully Discharged
- AC Input Supply Failure
- AC Input Under Voltage
- DC Under Voltage
- DC Over Voltage
- Thyristor/Diode Failure
- DC Earth Leakage
- AC Input Fuse Blown-off / MCCB Trip
- DC Output Fuse Blown-off / MCCB Trip
- DC Battery Fuse Blown-off / MCCB Trip
- Filter Capacitor Fuse Blown-off
- Short Circuit

- Reverse Polarity
- Battery Current Limiter

The annunciation system shall have audio-visual arrangement and shall incorporate acknowledge, reset and test push buttons. On occurrence of fault, the corresponding window lights up and stays in lighted condition until the fault is cleared and the reset push button is pressed. At least 20% spare annunciation windows with accessories shall be provided.

5.7. Grounding

All panels shall be equipped with an earth bus securely fixed inside the panels. Earth conductor shall be 25 x 6 mm Cu flat. When several panels are mounted adjoining one another, the earth bus shall be made continuous and necessary connectors and clamps for this purpose shall be included in the scope of supply. Provision shall be made to extend the earth busbar to future adjoining panels on either side.

Provision shall also be made on the earth busbar of the end panels for connecting the purchaser's grounding cable.

All metallic cases/frames of relays, instruments and other panel mounted equipment shall be connected to the earth bus by independent copper wires of minimum size 2.5 mm^2. Suitable clamp connectors shall be used. Soldering shall not be permitted. Green wires shall be used for earth connections.

5.8. Painting

Panels shall be painted with two coats of epoxy based primer after the pretreatment of surface like de-rusting, degreasing, descaling, etc., Subsequent to application of two coats of primer, stoved after each coat, final two coats of shade RAL 7035 shall be applied. The paint thickness shall not be less than 80 microns.

5.9. Test Requirements

5.9.1. Routine Test

All the equipment and components upon completion of manufacture shall be routine tested as stipulated in relevant Indian Standards. In case where Indian Standards are not available / such equipment shall be routine tested in line with equivalent reference standards.

Routine tests and verifications to be conducted on the charger panel are as listed below.

- Power frequency voltage dry test
- Voltage test on control auxiliary circuits
- Tests of the auxiliary electrical devices
- Verifications of wiring and functional test of all circuits
- Design and visual checks
- Voltage regulation check from 0 to 100% load with + 10% input voltage variation

- Ripple content measurement
- Routine tests on all component parts.

5.9.2. Type Test

Only type tested equipment shall be acceptable. All such type tests, which have not been conducted so far, shall be conducted during the tenure of contract by the Supplier without any cost and time implications. The purchaser reserves the right to witness such tests.

6. BATTERY

6.1. General

The battery shall be VRLA type, long life and sealed maintenance free type sized to meet the DC load requirements.

The battery shall be designed for maximum durability during all service conditions including high rate of discharge or over boost charge, accidental shorting, polarity reversals, rapid fluctuation of load and for severe shock & vibration.

The Battery shall normally be permanently connected to the load in parallel with battery charger and shall supply the load during emergency conditions when AC supply is lost.

DC system shall operate as ungrounded system.

Necessary ageing margin (at least 25%) on capacity shall be considered. Battery shall be adequately sized considering necessary design margin (at least 10%) and derating factors due to minimum and maximum ambient temperatures.

Battery shall be suitable for being boost charged to fully charged condition from fully discharged condition within 10 hours.

The battery shall be of high ampere-hour efficiency and suitable for indoor installation in humid and tropical atmosphere.

The batteries shall be sealed type and assembled in heat resistant shock absorbing containers with covers fixed in place to form a permanent leak proof seal. Explosion proof vent plugs shall be provided.

Terminal posts shall be of solid brass and adequately dimensioned for strong construction with least electrical resistance and complete with acid resisting connecting strips, bolts and nuts. Positive and negative terminal posts shall be clearly and indelibly marked for easy identification.

The cells shall be supported on porcelain insulators fixed on to the rack with adequate clearance between adjacent cells.

Lead-coated bent copper plate, tubular copper lugs, teak wood clamps, bolts, nuts, washers, etc., shall be furnished for connection of outgoing cables.

The batteries shall be furnished complete with cell inter-connectors, inter-row/inter-tier connectors and wooden racks.

The construction of the racks, unless otherwise mentioned shall be suitable for fixing to a flat concrete floor. The arrangement of the rack shall be such as to allow easy access with adequate space for normal maintenance in battery room.

The end cell voltage shall be selected so that it will suit the system requirement considering the full duration of duty cycle, normal operating voltage, voltage drop on cables and voltage variation for which DC consumers are designed.

6.2. Battery Isolation Box

Each battery bank shall be provided with individual local battery isolation box with adequate capacity of Fuse/MCCB for isolation.

The battery isolation box shall be provided with necessary fuse fail detection circuit and same shall be wired to respective charger circuit.

The battery isolation box shall be of metal enclosure, wall mounting, indoor type with front doors. The enclosure material shall be CRCA and thickness shall not be less than 2mm. The paint shade shall be RAL7035 and thickness shall not be less than 80 microns.

6.3. Accessories

Each Battery bank for the service indicated shall be complete with accessories and devices including but not limited to the following:

- Battery rack shall be of good quality, Teakwood and painted with 2 coats of approved acid resisting paint. The complete racks shall be suitable for being bolted end to end to form a continuous row.
- Set of inter cell, inter-tier and interbank connectors as required for the complete installation.
- One (1) -3- 0- +3 V DC voltmeter with suitable leads for measuring cell voltage.
- Numbering tags for each cell shall be attached on to the racks.
- Suitable set of spanners, assembly wrenches, Thermometer with plug and cap.

6.4. Tests

6.4.1. Routine Test

The battery and its associated components upon completion of manufacture shall be subject to routine & acceptance tests as stipulated in relevant Indian Standard. In case where Indian Standards are not available / applicable such equipment shall be routine tested in line with equivalent reference standard.

6.4.2. Type Test

Type test (on at least four random samples) shall be carried out and test certificates shall be furnished. Only type tested equipment shall be acceptable. All such type tests, which have not been conducted so far, shall be conducted during the tenure of contract by the Supplier without any cost and time implications. The purchaser reserves the right to witness such tests.

All acceptance tests as per IEC shall be carried out at site.

7. DC DISTRIBUTION BOARD

7.1. Constructions

The DC Distribution board shall be formed using distinct vertical sections each comprising the following compartments:

- Two numbers of Incomer breaker compartment and one number of bus coupler breaker with mechanical key interlock for the connected DC loads and its control chamber
- A completely metal enclosed horizontal busbar compartment running horizontally at top.
- Completely enclosed vertical bus bars serving all outgoing feeder modules in the vertical section.
- Vertical compartments - each vertical compartment shall have outgoing feeders (MCCB) for DC motor starter panels and outgoing MCB feeders as per the DC distribution SLD.
- Power cable chamber for incomer & outgoing feeders in separate vertical sections.

DCDB shall be indoor, dust, damp & vermin proof construction, totally enclosed compartmentalized, floor mounted, self-standing type with access from front by means of hinged doors. The panel shall allow extension at either end. The enclosure protection of the panel shall be IP54 protection.

The framework of the board shall be constructed of performed steel channels, angles and side sheets bolted together and suitably reinforced to form a rigid, self-supporting, compact assembly to function properly under both normal and short circuit conditions.

The panel shall have bottom cable entry for the incoming and outgoing feeders.

Name plates of elegant and durable design and quality shall be fixed to the individual compartments door. Indicator labels shall also be provided for all control devices and components inside the panel. Inscription plates provided shall be engraved on black anodized aluminium.

Cut outs shall be true in shape and free from sharp edges.

The panel shall be provided with metal base channel minimum 75mm in height for facilitating grouting on the floor.

Cold Rolled Sheet steel of 2.0mm thickness shall be used for all members except for partitions where in may be of 1.6mm.

All doors and removable covers shall have neoprene gaskets. All the doors for the panel in the front shall have individual sheet steel hinged doors with concealed hinges and shall be earthed with 1.5mm2 copper flexible wire of green colour.

Removable undrilled gland plates of 3 mm thick sheet shall be provided at the bottom of the DCDB.

All live parts shall be suitably shrouded with non-deteriorating insulating barriers so as to prevent any accidental contact with these by the operating/maintenance personnel while the panel is in service and the modules are opened for inspection/servicing etc.

The switchboards shall have maximium height of 2115 mm. Maximum operating height of device on the panel shall not exceed 1800mm and minimum shall not be less than 400mm. However, height of all the panels shall be uniform.

At least 400 mm free space (or more for multiple runs of large power cable sizes) shall be kept from gland plate at panel bottom up to the point where incoming / outgoing cables are going to be terminated. This is for ensuring correct bending radius of the incoming / outgoing cables while terminating them.

7.2. Incomer & Bus Coupler Module

The incomer and buscoupler module shall be provided with suitably rated MCCB's.

The MCCB shall have microprocessor based Over current, Short Circuit and Earth fault protection releases.

The incomer & bus coupler modules shall be of approved quality and make and comprise of the following:

- MCCB
- MCB for Control Circuits
- Selector Switches
- Set of cluster type LED lamps equipped with Low Voltage Glow Protection for ON / OFF / TRIP
- DC Voltmeters & Ammeters

- DC shunt for Ammeters
- Terminal blocks, interconnection bars / wires, ferrules, etc. as per requirement

7.3. Bus Bars

The busbars shall be of hard drawn high conductivity electrolytic grade (63401 WP) Aluminium conforming to IS: 5082. The busbars shall be of uniform cross section along its entire length. No tapering of the busbar cross section is allowed.

The busbars shall be designed for maximum current density of 0.8 A/mm2 for Aluminium bus bars and 1.2 A/mm2 for copper busbars.

Horizontal and vertical busbars shall be provided with FR PVC sleeves of high quality and adequately shrouded to prevent accidental contact. The sleeve used shall be of the heat shrinkable type.

The switchboard shall be provided with power busbars for DC +Ve & -Ve terminals. The Busbars fed by the incomer shall run continuously throughout the length of the switchboard.

The cross section of the busbars shall be uniform throughout after considering all derating factor.

The necessary calculation for selecting size of bus bar and insulator shall be furnished during detailed engineering for owner/consultant approval.

Power shall be distributed to each outgoing feeders module or compartment by a set of vertical busbars. The vertical busbars shall run behind the modules. The busbars shall run horizontally in a separate enclosure but may run in an enclosure located within the power busbar compartment.

The busbars shall be phase identified by colour, throughout the length or identification of busbars at regular intervals. Colour code shall be RED & Black for DC buses. Busbars shall be rated for 35° C temperature rise over an ambient of 50° C.

Busbars shall be supported on tough, non-hygroscopic, resin bonded self extinguishing fire retardant insulators of SMC / DMC with ribbed construction to prevent tracking due to dust accumulation and to have larger creepage paths. All the busbars shall be individually supported. If a common support is provided for DC bus, anti-tracking barriers shall be incorporated. Busbars and supports shall withstand the maximum stresses that are likely to be induced by the short circuit currents specified.

The joints in the busbars shall be of the bolted type and it shall be ensured that the following precautions are observed:

- The contact pressure must be ample and this shall be maintained during the time the panel is in service.

- The surfaces of the conductors must be clean.
- With flat conductors, the overlap should be equal to or greater than the width of the bars or ten times the bar thickness whichever is greater.
- The joints shall be treated by the application of joint compound to render the joint moisture-proof.
- For the joints, use bolts of cadmium plated high tension MS bolts and nuts.

All busbar joints shall be of the bolted type. Ring washers shall be used for joints to prevent loosening of the nuts and over heating. Lock nuts shall be incorporated for the bolts besides the main nuts. The bolts and nuts used in the current carrying paths shall all be of steel cadmium plated and the hardware shall have adequate cover of the protective cadmium coating to obviate corrosion while the board is in service. Proper work space is required for tightening and checking the bolts and joints.

The busbar chambers at the top shall be totally segregated from the rest of compartments and rendered totally inaccessible under normal circumstances. No equipment / Instrument shall be mounted on the busbar chamber.

7.4. Outgoing Feeder Module

The outgoing feeder modules for DC starter panels and its spare feeder shall be provided as individual cubicle.

The outgoing feeders to DC starter panel shall have the following components;

- MCCB
- MCB for Control Circuits
- Set of cluster type LED lamps equipped with Low Voltage Glow Protection for ON / OFF / TRIP
- Terminal blocks, interconnection bars / wires, ferrules, etc. as per requirement

The outgoing feeders to STG control panels and MV switchgear panels shall have the following components;

- Individual MCB's with adeuate rating
- Terminal blocks, interconnection bars / wires, ferrules, etc. as per requirement

7.5. Grounding

All panels shall be equipped with an earth bus securely fixed inside the panels. Earth conductor shall be 25 x 6 mm Cu flat. When several panels are mounted adjoining one

another, the earth bus shall be made continuous and necessary connectors and clamps for this purpose shall be included in the scope of supply. Provision shall be made to extend the earth busbar to future adjoining panels on either side.

Provision shall also be made on the earth busbar of the end panels for connecting the purchaser's grounding cable.

All metallic cases/frames of relays, instruments and other panel mounted equipment shall be connected to the earth bus by independent copper wires of minimum size 2.5 mm^2. Suitable clamp connectors shall be used. Soldering shall not be permitted. Green wires shall be used for earth connections.

7.6. Painting

Panels shall be painted with two coats of epoxy based primer after the pretreatment of surface like de-rusting, degreasing, descaling, etc., Subsequent to application of two coats of primer, stoved after each coat, final two coats of shade RAL 7035 shall be applied. The paint thickness shall not be less than 80 microns.

8. SPARES

Supplier shall furnish a list of recommended spares required for two years of trouble free normal operation. Commissioning spares required if any, shall be provided.

9. DRAWINGS, DATA AND MANUALS

To be submitted with the bid

- Battery layout drawings with dimensions
- Battery cell voltage characteristics and data for different discharge rates
- Technical leaflets
- Battery sizing calculations as per IEEE 485.
- Type test certificate for similar rating

The following drawings / documents shall be submitted with four (4) copies for approval within one week after placement of order.

- GA drawing of battery and battery charger.
- Foundation drawing of battery and battery charger.
- Technical data sheet.
- Schematic and wiring diagram of battery charger.
- Write-up on operation and control philosophy of battery charger.
- Commissioning, Operation and maintenance instructions of battery & battery charger

It is to be noted that, nothing in this specification shall be construed to relieve the supplier of their responsibility for design, manufacture, testing and performance of the 220V VRLA battery & charger.

10. ANNEXURE

- Annexure-I Scope of Supply
- Annexure-II DC System GTP
- Annexure-III Battery Charger & Battery Sizing Calculation
- Annexure-IV DC Distribution SLD